TAK RIG

USER GUIDE

v0.19.3

Offline navigation, convoy awareness,
field communications, and connected data
for off-road and overland travel.

Designed for landscape Android tablets and vehicle displays

Main Features Guide | October 2026

TAK RIG AT A GLANCE

This guide covers the main end-user workflows in TAK Rig v0.19.3.

TAK Rig is built for landscape Android tablets and vehicle-mounted Android systems used for off-road and overland travel. It combines offline navigation with convoy awareness, local data sharing, communications, terrain tools, and field information in one interface.

VERSION NOTE

This guide describes TAK Rig v0.19.3. Later releases may move controls or add capabilities. The release notes on GitHub should be treated as the final reference for version-specific changes.

Offline-first navigation

Prepare maps, routing, terrain, and regional data before leaving coverage.

Convoy awareness

Share positions and trip data between compatible TAK Rig systems.

Route intelligence

Record drives and add trail conditions, hazards, gates, water crossings, camps, recovery points, and notes.

Recent satellite imagery

Use Sentinel-2 imagery to supplement older basemaps when connectivity is available.

Direct data transfer

Move supported offline datasets between TAK Rig devices over local connections.

Vehicle-first UI

The complete TAK Rig interface runs directly on a large landscape Android display.

Guide at a glance

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1 QUICK START

A practical pre-trip checklist for a newly installed TAK Rig system.

1. Install the current TAK Rig APK over the existing installation when updating. Normal updates are intended to preserve app data and settings.
2. Allow location access when prompted. TAK Rig uses location for self-position, navigation, route recording, convoy position sharing, and other location-aware tools.
3. Open Basemaps and choose the map you want to use. Add or import any offline maps you plan to rely on.
4. Open Settings > DATA. Set the Navigation Data Manager (NDM) endpoint if you use one, then install the regional packages you need for routing and terrain.
5. Open Sources and confirm your GPS source is healthy. Enable any optional sources you want, such as Weather, Traffic, Cellular Coverage, ADS-B, or Gateway Data.
6. If traveling with other TAK Rig users, configure your preferred sharing/communications method and verify that peers can see one another before leaving coverage.
7. Create a Settings Backup before a major trip or before making large configuration changes.

> **BEFORE YOU LEAVE COVERAGE**
> Open the maps and regions you plan to use, verify offline routing, verify terrain data if you use 3D, and test the exact convoy/network setup you will use in the field.

Recommended trip-prep workflow

MAPS
Download or import the basemap coverage you need.

DATA
Install routing, 3D terrain, cellular, and outdoors packages.

ROUTES
Save planned routes and important waypoints.

SOURCES
Verify GPS and optional live overlays.

CONVOY
Confirm position sharing, Chat, Data Sync, and/or Voice.

BACKUP
Save your settings configuration before departure.

2 INTERFACE TOUR

The map stays central; most tools open as panels over it.

TAK Rig is designed around a large landscape display. The map is the main working area, while the left rail provides quick access to the primary tools. The rail can scroll when every tool does not fit on screen.

Rail Tool	Main Use
Navigate	Start navigation to a selected destination.
Basemaps	Choose the base map shown underneath your overlays.
Chat	Read and send supported messages.
Data Sync	Work with shared Missions and synchronized map data.
Marker	Create a point/waypoint using the icon registry.
Shapes	Create rectangles, circles, ellipses, polygons, and freehand areas.
Record Route	Record your driven path and trail observations.
Routes	Create, review, edit, and use saved routes.
Lasso Select	Select map objects inside a drawn area.
Measure	Measure distance and create reusable geometry.
Sources	Control GPS inputs and optional live data overlays.

BASEMAPS VS SOURCES

Basemaps change the map underneath everything else. Sources control live or optional information layered on top, such as weather, traffic, aircraft, cellular coverage, or external gateway feeds.

Settings tabs

Open Settings for persistent app configuration. The main tabs are GENERAL, MAPS, DATA, TRANSFER, VEHICLE, TAK (when enabled), and VOICE. This guide focuses on the non-TAK settings used for off-road and overland workflows.

2A SETTINGS: UNITS & TAK TAB

Customize measurement display and reveal the advanced TAK configuration tab when needed.

Units in GENERAL settings

Open Settings > GENERAL and scroll to Units. TAK Rig defaults to Imperial (US) on new and upgraded installations. Changes apply to TAK Rig display and measurement-entry fields immediately; underlying routing, GPS, terrain, CoT, and saved geometry remain in their canonical internal units.

UNIT SYSTEM PRESETS

Imperial (US) - sets the full group to US customary units.
Metric - switches the full group to metric units.
Custom - appears automatically when you change any individual category.

Category	Choices
Temperature	Fahrenheit (F) / Celsius (C)
Speed	mph / km/h / knots
Distance	miles / kilometers
Short Distance / Accuracy	feet / meters
Altitude / Elevation	feet / meters
Vertical Speed	ft/min / m/s
Wind Speed	mph / km/h / knots
Precipitation	inches / millimeters
Area	sq ft, acres, sq mi / m2, hectares, km2

Unlock the TAK tab

The TAK tab is intentionally hidden from the normal Settings row. Use the GENERAL tab as the unlock control:

1. Open Settings and select GENERAL.
2. Tap the GENERAL tab 8 times within 10 seconds. GENERAL remains tappable while selected so the sequence can be completed.
3. TAK appears immediately before VOICE. The unlocked state persists across normal app restarts and updates.
4. To hide TAK again, repeat the same 8-tap sequence on GENERAL.

BACKUP & RESTORE

Unit selections are non-TAK settings and are included automatically in TAK Rig Settings Backup & Restore. TAK-tab settings remain excluded.

3 MAPS & SOURCES

Build the map view you want for the current part of the trip.

Basemaps

Use Basemaps to switch the primary map layer. TAK Rig can work with built-in and imported map sources, including offline map sources prepared before the trip.

- Choose a default basemap in Settings > MAPS.
- Speed-based basemap switching can be enabled when you want different map styles at different driving speeds.
- Imported maps can remain available offline when their source data is stored locally.
- Outdoors styling and terrain-related presentation can be adjusted in Settings > MAPS.

Sources panel

Sources lets you add or remove information without changing the basemap. Current source groups include GPS inputs, Weather, Traffic, ADS-B, Cellular Coverage, and Gateway Data.

Source	What it adds
GPS Inputs	Shows available location inputs, their status, accuracy, age, contribution, and rejected fixes.
Weather	Displays weather information when a configured source and connectivity are available.
Traffic	Adds TomTom traffic information when configured.
ADS-B	Displays supported aircraft tracks and details.
Cellular Coverage	Overlays installed cellular-coverage data by carrier with adjustable opacity.
Gateway Data	Displays configured external data feeds supplied through a TAK Gateway source.

TRAFFIC API KEY

The TomTom API key used by Traffic/Places is included in TAK Rig Settings Backup and is re-encrypted when restored to another installation.

4 OFFLINE DATA

Prepare regional data before you lose connectivity.

TAK Rig separates the visible map from several other offline datasets. Settings > DATA is the main place to manage those packages.

Navigation Data Manager (NDM)

If you use a TAK Rig Navigation Data Manager, enter its endpoint in Settings > DATA. TAK Rig can check the catalog, download supported packages, show progress, and keep multiple regions installed at the same time.

Navigation

Local routing data used for road, unified, and off-road route calculation.

3D Terrain

Elevation data used by the 3D terrain engine and terrain-aware tools.

Cellular Coverage

Regional coverage data for the cellular overlay.

Outdoors Maps

Additional regional map data used by the Outdoors map experience.

Data Vault

The Data Vault settings can be used to control where selected large payloads are persisted. This is useful on systems where internal storage is limited or where you want offline data separated from the app itself.

TEST OFFLINE ROUTING

Before a remote trip, temporarily disable network access and calculate a route inside every region you expect to use. This confirms the routing package is really available locally.

5 NAVIGATION

Use installed local routing data for on-road, mixed, or off-road travel.

The Navigate tool starts route guidance to a selected destination. Destinations can come from map points, saved objects, or a location selected directly on the map.

Routing profiles

Profile	Best fit
Unified	General navigation where TAK Rig can combine available routing options.
Road	Road-focused routing using installed road navigation data.
Off-Road	Routing intended for off-road/trail-oriented route creation and use.

During navigation

- The active route remains visible on the map while guidance is running.
- TAK Rig can recalculate after a missed turn without immediately removing the last working route.
- Multiple installed regional packages can remain available; TAK Rig selects suitable local routing data for the requested trip.
- A saved route can also be traversed when you want to follow an existing track rather than calculate a new destination route.

> **ROUTING COVERAGE**
> If TAK Rig cannot calculate a route, first confirm that the start and destination are both covered by an installed navigation package.

6 ROUTES & RECORDING

Plan ahead, record what you actually drove, and capture trail conditions as you go.

Create a route

1. Open Routes and choose Create Route.
2. Choose Road or Off-Road for the new route.
3. Add the route points you want to travel through.
4. Review the result, edit the geometry if needed, then save the route for later use.

Saved routes can be reviewed, edited, used for traversal, shared through supported Data Sync workflows, and displayed with route statistics/elevation information.

Record a route

1. Open Record Route and wait for a valid GPS fix.
2. Choose the route type that best matches the trip: Drive, Offroad, or Overlanding.
3. Start recording. TAK Rig adds GPS samples as you travel.
4. Add notes or trail-condition annotations when something worth remembering occurs.
5. Stop the recording, give the route a useful name, and save it. You can also discard an unwanted recording.

> **GPS PROTECTION IN V0.19.3**
> TAK Rig now quarantines implausible one-frame location jumps before they replace the trusted live position. The route recorder also rejects any individual segment at or above 3 km so a GPS teleport is not saved into the route geometry.

6A TRAIL ANNOTATIONS

Turn a recorded route into a useful field log instead of just a line on the map.

While recording, TAK Rig can attach observations to the current location or to a recent segment of the route. Segment annotations can be applied to the last 1/4 mile, 1/2 mile, 1 mile, 2+ miles, or 5+ miles.

Category	Examples
Surface	Paved, Gravel, Dirt, Sand, Rock, Mud, Snow / Ice, Mixed
Difficulty	Easy, Moderate, Difficult, Extreme
Road Condition	Smooth, Washboard, Rutted, Eroded, Loose, Muddy, Deep Sand, Flooded
Passability	Passenger Car, AWD Recommended, High Clearance, 4x4 Recommended/Required, No Trailers
Hazard / Obstacle	Drop-off, low clearance, rock, tree, debris, washout, other
Gate / Access	Open, Closed, Locked, Seasonal, Private, Unknown
Water Crossing	Dry, Shallow, Moderate, Deep, Impassable
Fuel / Service	Fuel, Service, Repair, Air, Other
Camp / Stop	Camp, Rest Stop, Overnight, Staging, Parking
Recovery	Recovery Point, Winch Point, Turnaround, Bypass, Safe Pull-off
Comms / Cell	Cell Service, No Cell Service, Good/Poor Radio, Emergency Comms
Note / Scenic	Freeform note or scenic point

WHY ANNOTATE

A recorded route becomes much more useful on the next trip when it tells you where the washboard started, where the locked gate was, whether the water crossing was passable, and where you had cell or radio coverage.

7 3D TERRAIN

Use terrain context to understand the route, not just the line.

TAK Rig includes a dedicated 3D terrain view that can be used as a free map or while following your own position. It requires compatible terrain data for the area being viewed.

FOLLOW

Keep your current GPS position as the center of the 3D camera.

ROTATE

Orbit the view around the current map/vehicle focus.

PITCH

Change the viewing angle from shallow to more top-down.

ZOOM

Change the camera distance while keeping terrain context.

MOVE

Move the 3D map manually; manual movement exits Follow.

EXAGGERATION

Increase or reduce terrain relief in Settings > MAPS.

Routes, points, and your own vehicle marker can be shown in perspective against the terrain. TAK Rig progressively refines terrain and imagery as higher-detail content becomes available.

IF 3D LOOKS FLAT OR EMPTY

Check Settings > DATA and confirm the correct 3D Terrain package is installed for the region.

8 MAP TOOLS

Mark places, draw areas, measure distance, and organize what matters on the trail.

Points / Markers

Use Marker to place a point on the map. Choose an icon from the icon registry, give the point a useful name, and use it for camps, fuel, trailheads, hazards, meet-up points, recovery locations, or anything else you want to keep visible.

Shapes

Shapes can be used to outline an area rather than a single point. TAK Rig supports rectangles, circles, ellipses, polygons, and freehand shapes. Shape size, rotation, stroke, fill, opacity, and other presentation options can be adjusted.

Measure

Use Measure for quick distance checks on the map. A closed measurement can also be turned into a saved route when that geometry is useful for planning.

Lasso Select

Lasso Select makes it easier to work with multiple objects in one map area without tapping each item individually.

Geofences

Points, routes, and shapes can be used as geofence sources for location-aware alerts and boundary monitoring. Configure geofences only after confirming the underlying object geometry is correct.

9 CONVOY AWARENESS & DATA SYNC

Keep the group aware of positions and shared trip information.

TAK Rig can exchange position and map information with other compatible systems using supported TAK or mesh-oriented transports. The exact network setup depends on how your group is connected.

Position sharing

See compatible group members as map objects when their position is being shared.

Chat

Keep supported conversations alongside the map instead of switching to a separate app.

Data Sync

Share points, routes, shapes, and files through Missions.

Token Mesh

Use an encrypted shared-token mesh mode for supported local/overlay network paths.

Mission organization

Keep trip data grouped instead of mixing every shared object into one unstructured map.

Optional TAK connectivity

Connect to compatible TAK infrastructure when your setup requires it.

CONVOY SETUP RULE

Test the exact peer/network configuration while all vehicles are together. Do not wait until the group is already separated and out of coverage to discover that one device is using a different network or sharing mode.

Protected Missions

Missions can be protected so access is not granted simply by seeing the Mission name. TAK Rig can use Mission passwords and token-based unlock workflows, including a QR token for transferring access to another TAK Rig device.

TREAT MISSION TOKENS LIKE PASSWORDS

A Mission unlock token grants access to that protected Mission. Share it only with people who should have access.

Hardware & Network Options

TAK Rig sits on top of the network your vehicles provide. You are not locked to one radio or one type of connection; choose the network that fits the trip and the distance between vehicles.

Standard Wi-Fi

Use a vehicle hotspot, travel router, or shared Wi-Fi network when the group is staying relatively close together.

Wi-Fi HaLow

802.11ah uses sub-GHz spectrum for longer-range IP networking and can be a strong fit for convoy and remote-area links.

Cellular / Internet

When coverage exists, normal internet connectivity can support remote services, servers, and long-distance links.

Virtual / Mesh Networks

Overlay networks such as ZeroTier, or compatible IP-capable mesh/radio systems, can connect vehicles when the underlying hardware supports them.

SUGGESTED HALOW HARDWARE - HELTEC HT-HD01 V2

For a practical TAK Rig HaLow setup, the Heltec HT-HD01 V2 is a good starting option. A paired AP + STA set creates a HaLow link while the Android display can connect to the dongle over 2.4 GHz Wi-Fi, or USB-C where supported. Heltec lists 802.11ah operation in the 902-928 MHz band, up to 27 +/- 1 dBm transmit power on V2, and up to 32.5 Mbps at 8 MHz. The manufacturer describes typical communication range in the 1-2 km class; actual range depends heavily on antennas, mounting, terrain, interference, and line of sight.

ABOUT THE 10-MILE HALOW TEST

The linked Morse Micro field test in Joshua Tree demonstrated a Wi-Fi HaLow connection at about 15.9 km (10 miles) with roughly 2 Mbps UDP throughput under near-ideal, low-interference line-of-sight conditions. The test used Morse Micro evaluation hardware, not the Heltec HT-HD01, so it should be viewed as a demonstration of what 802.11ah can achieve under favorable conditions - not as a guaranteed range for a TAK Rig vehicle installation.

HaLow test video: https://www.youtube.com/watch?v=fBMgZah2Z7g

Heltec HT-HD01 V2: https://heltec.org/project/ht-hd01/

10 DEVICE-TO-DEVICE TRANSFER

Move large offline datasets locally instead of downloading the same package in every vehicle.

Settings > TRANSFER is designed for direct TAK Rig-to-TAK Rig dataset transfer. Both devices should open the Transfer tab during discovery.

Transfer capability	Typical use
Navigation	Share prepared regional routing data.
Elevation	Share terrain/elevation data.
Cell Coverage	Share regional cellular-coverage data.
Basemap	Share supported basemap information.
Offline Map	Share supported offline map datasets.

1. Open Settings > TRANSFER on both TAK Rig devices.
2. Allow the devices to discover one another. A shared LAN is used automatically when available; Wi-Fi Direct can be used when there is no shared network.
3. Choose the destination device and the dataset you want to send.
4. Accept the incoming transfer on the receiving device.
5. Leave the transfer running until TAK Rig reports completion. Accepted transfers can continue even if you leave the Transfer tab.

> **WHY THIS MATTERS**
> For a convoy, one fully prepared system can become the local source for large regional packages when internet access is slow, expensive, or unavailable.

11 RECENT SATELLITE IMAGERY

Supplement static basemaps with Sentinel-2 imagery when connectivity is available.

TAK Rig can request Sentinel-2 imagery for the area being viewed. The imagery is useful when you want a newer look at terrain or land conditions than an older satellite basemap can provide.

AUTO date

TAK Rig looks through qualifying recent scenes and continues past no-data results.

Fixed date

Choose a specific imagery date when you need a known acquisition.

Cloud filtering

Use the maximum cloud setting to avoid poor scenes.

Local cache

Downloaded imagery is cached so it does not need to be fetched every time.

Diagnostics

Settings > MAPS can test OAuth, Catalog, and Process API stages separately.

Date awareness

TAK Rig displays the imagery date so you know what you are looking at.

Typical setup

1. Enter your Copernicus/Sentinel credentials in Settings > MAPS.
2. Set the maximum cloud value you are comfortable with.
3. Use TEST SENTINEL-2 if you want to confirm authentication and imagery services before a trip.
4. Choose Sentinel imagery as needed while connected, then allow the relevant areas to cache before going offline.

> **IMAGERY IS NOT LIVE**
> Sentinel-2 is recent satellite imagery, not a real-time camera feed. Acquisition dates and cloud conditions vary by area.

12 VOICE & LIVE SOURCES

Add communications and live information when your hardware/network setup supports it.

Voice

Settings > VOICE controls TAK Rig voice communication. Available behavior depends on the selected network/transport and the audio devices connected to the Android system. Use the Voice status area to confirm whether the selected input, output, and network mode are ready.

> **TEST AUDIO BEFORE DEPARTURE**
> Verify the actual microphone, speakers, push-to-talk method, and network path you will use in the vehicle.

Weather and Traffic

Enable Weather or Traffic from Sources when connectivity is available. Traffic uses the configured TomTom API key; Places search can use the same credential.

ADS-B

The ADS-B source can display supported aircraft tracks on the map. Aircraft details can include registration/ICAO, model/type, altitude, speed, heading, squawk, and other available metadata. ADS-B is an optional source and starts disabled each TAK Rig process.

Cellular Coverage

When regional cellular data is installed, the Sources panel can overlay coverage by carrier. Use the carrier controls and opacity setting to compare expected coverage with the route ahead.

Gateway Data

Gateway Data can bring configured external feeds into the map through a compatible TAK Gateway setup. The exact feeds depend on your gateway configuration.

13 SETTINGS BACKUP & RESTORE

Carry your TAK Rig configuration forward without bundling large map and routing payloads.

Settings > GENERAL includes BACKUP SETTINGS and RESTORE SETTINGS. The v0.19.2+ backup architecture is designed to capture persistent user-configurable settings outside the TAK tab, including newly added non-TAK settings by default.

Included

- Map, navigation, source, route-presentation, vehicle, voice, data-vault, transfer-related and other persistent non-TAK user settings.
- Portable non-TAK credentials such as Sentinel/Copernicus and TomTom, re-encrypted for the destination installation during restore.
- Supported XTAK Mesh Voice channel credentials through the portable credential path.

Not included

- TAK-tab identity, server, certificate, and TAK credential settings.
- Large downloaded payloads such as offline map tiles, navigation graphs, terrain packages, cellular packages, and cached satellite imagery.
- Runtime state, caches, download progress, diagnostics, and other non-setting data.

> **BACKUPS CAN CONTAIN CREDENTIALS**
> Treat a TAK Rig settings backup as sensitive. Store it somewhere you trust and avoid posting it publicly.

Good times to create a backup

- Before a major trip.
- Before changing many map/source settings.
- Before reinstalling or moving TAK Rig to another compatible device.
- After you finish configuring a known-good vehicle setup.

14 FIELD TIPS & TROUBLESHOOTING

The fastest checks for the problems most likely to matter away from home.

Symptom	What to check first
GPS marker jumps or freezes	Open Sources > GPS. Check source age, accuracy, contribution, and rejected-fix status. Give TAK Rig time to reacquire a stable fix.
Route cannot be calculated	Confirm the required navigation package is installed and covers both the start and destination.
3D terrain is missing/flat	Confirm the regional 3D Terrain package is installed in Settings > DATA.
Offline map is blank	Switch basemaps and confirm the chosen source actually has local coverage for the current zoom/area.
Peer is not visible	Confirm both devices are on the intended network/mesh mode and sharing is enabled. Test discovery while devices are together.
Transfer peer not found	Open Settings > TRANSFER on both devices and confirm LAN or Wi-Fi Direct connectivity.
Sentinel imagery fails	Run TEST SENTINEL-2 in Settings > MAPS and read which stage failed: OAuth, Catalog, or Process API.
Traffic/Places unavailable	Confirm connectivity and verify the TomTom API key in the Traffic source configuration.
Voice not ready	Check Settings > VOICE for selected input/output device and network status.
Storage is getting tight	Review downloaded regional data and Data Vault choices; maps, terrain, and routing packages can be large.

GPS FIELD NOTE

v0.19.3 keeps the last trusted position active while a suspicious jump is being checked. If several consistent fixes confirm that you truly relocated, TAK Rig can deliberately re-anchor to the new position.

Pre-trip five-minute test

1. Confirm GPS is stable.
2. Calculate an offline route.
3. Open 3D terrain for the trip area.
4. Open the exact offline basemap you plan to use.
5. Confirm convoy peers / Chat / Voice as applicable.
6. Create a fresh settings backup.

QR QUICK REFERENCE

A compact reminder of where the most-used functions live.

Task	Where to go
Change base map	Basemaps
Start destination navigation	Navigate
Create a waypoint	Marker
Create an area	Shapes
Record the current drive	Record Route
Review or create saved routes	Routes
Measure distance	Measure
See weather / traffic / ADS-B / cellular	Sources
See GPS input health	Sources > GPS
Share Missions / trip objects	Data Sync
Send messages	Chat
Install routing / terrain / cellular data	Settings > DATA
Send offline datasets to another TAK Rig	Settings > TRANSFER
Configure terrain exaggeration / Sentinel	Settings > MAPS
Configure voice	Settings > VOICE
Backup settings	Settings > GENERAL > BACKUP SETTINGS

CURRENT DISPLAY TARGET

TAK Rig v0.19.3 is optimized for larger landscape Android tablets and vehicle-mounted Android systems. The current layout is not intended as a compact handset interface.

Safety and use

Off-road and backcountry travel can involve closures, private property, weather, terrain hazards, inaccurate map data, GPS errors, mechanical problems, and areas without communications coverage. Use TAK Rig as one tool in a larger preparation plan. Carry appropriate maps, recovery equipment, communications, and backup navigation methods for the environment you are entering.

TAK Rig is under active development and should not be treated as a certified navigation or emergency-communications system.

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